



AI City Architect



人工智慧城
市建築師

Programme Summary

AI City Architect is a project-based, age-differentiated curriculum (P1–P6, ages 6–11) that introduces students to artificial intelligence through the lens of urban systems. Across 20 sequential lessons, learners explore how AI can manage and optimize real-world city infrastructure from waste sorting and traffic flow to power grids, water networks, healthcare, air traffic, finance, and civic sentiment analysis.

Each lesson presents a core urban challenge, with students progressively mastering AI concepts such as object detection, pathfinding, swarm intelligence, predictive modeling, semantic segmentation, reinforcement learning, and resource tokenomics. The curriculum culminates in three modules where students design, optimize, and simulate their own AI-driven city, applying previous skills while managing budgets, long-term macro-trends, and unexpected crises (e.g., heatwaves, economic shocks).

The course emphasizes hands-on digital simulations, drag-and-drop coding, data annotation, model training, and strategic decision-making, building technical AI literacy and systems thinking.

課程簡介 (中文)

AI 城市規劃師 是一套專為 6 至 11 歲學生 (P1–P6) 設計、分齡且專案導向的人工智慧課程。透過 20 個循序漸進的單元，學生將學習如何運用 AI 管理並最佳化真實的城市基礎設施——從廢物分類、交通流量、電網、水資源網路，到醫療保健、飛航管制、財務稽核與市民情緒分析。

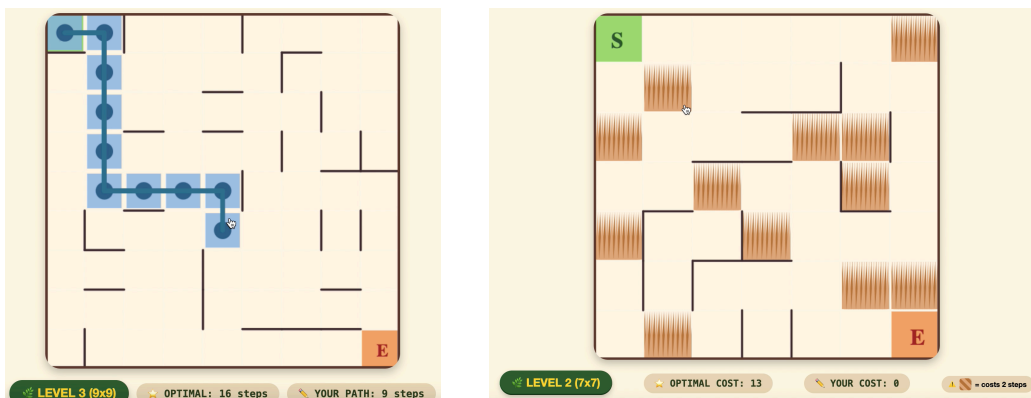
每個單元聚焦一個特定的城市挑戰，學生逐步掌握物件偵測、路徑規劃、群體智慧、預測模型、語義分割、強化學習及資源代幣經濟學等 AI 核心概念。課程最後以三個總整單元收尾：學生需設計、優化並即時模擬自己打造的 AI 城市，應用先前所學的技能，同時管理預算、長期宏觀趨勢（如人口老化、房價變動）以及突發危機（如熱浪、經濟衝擊）。本課程強調數位模擬、拖曳式程式編寫、資料標註、模型訓練與策略決策，全面培養學生的 AI 素養與系統思維能力。

Sample Lesson (#4, P3)

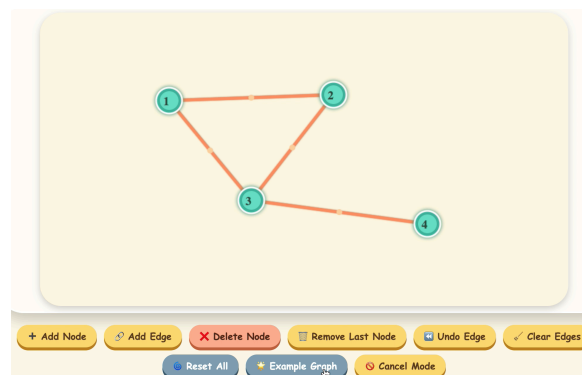
Led by the guide Nova, this interactive lesson introduces students to path optimization. Through gamified challenges, students explore how to find the fastest, most efficient route to a destination, a foundational skill in computer science and logistics.

Core Concepts Covered

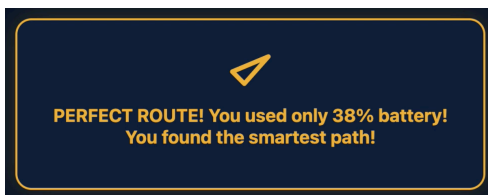
- **Grid Navigation:** Students solve grid challenges to find the shortest path from start to end.
- **Variable Terrain Costs:** The curriculum introduces squares that count as two steps instead of one, teaching students to calculate cumulative path costs rather than just simple distance.



- **Graph Theory Basics:** Nodes defined as objects or points in a network, and edges as the lines that connect two nodes together.



- **Weighted Graphs:** Students learn that edges can have different costs or times (e.g., an uphill road taking longer). The final challenge requires finding a route with the smallest sum of weights to conserve a drone's battery.



Real-World Applications: abstract graph structures are mapped directly onto everyday systems to ensure learning is tangible.

- **Physical Road Networks:** Representing landmarks (such as a Home, Grocery Store, and School) as nodes, and the literal streets connecting them as edges.
- **Social Networks:** Using nodes to represent individuals and edges to explicitly map out relationships, showing how people in a community know each other.

Key Learning Takeaway

The lesson concludes by highlighting computational complexity, and explaining how AI can help. Students experience firsthand that as a network grows, finding an optimal path becomes a major challenge because they must evaluate a rapidly increasing number of possible combinations.

Programme Overview

Lesson 1: Intelligent Waste Sorting

Level	Description
P1 (Age 6)	Students engage in an interactive digital sorting exercise, categorizing waste into appropriate recycling and general waste bins via drag-and-drop mechanics. They also participate in a progressive reveal guessing game to identify recyclable objects.
P2 (Age 7)	Participants classify objects based on complex, non-visual attributes such as weight, material composition, and texture. They also match descriptive clue cards with corresponding object photographs.
P3 (Age 8)	Using tablets, students annotate images by circling distinct features to train an AI model. They learn object detection (bounding boxes), geometry/landmarks (identifying corners and ridges), surface texture analysis, and text/symbol recognition (barcodes and logos).
P4 (Age 9)	Building on previous annotation skills, students prompt an AI to generate images based on specific feature combinations. They also engage in a riddle-based deduction game, identifying objects solely from their described characteristics.
P5 (Age 10)	Students apply the feature-annotation game specifically to recyclable materials, uploading their labeled datasets to train a foundational AI classification model.

P6 (Age 11)	Students refine their AI classification models using recyclable materials and are introduced to the critical machine learning concept of model overfitting.
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Lesson 2: Subsurface AI Scanning & Site Analysis

Level	Description
P1 (Age 6)	An AI survey vehicle maps a 2D land grid to identify subsurface hazards (e.g., pipes, artifacts, groundwater). The detector signals hazards, marking "No-Build Zones," allowing students to drill and investigate.
P2 (Age 7)	Students operate the scanner in two modes: a rapid sweep that misses deep, narrow hazards, and a slow, high-frequency pulse that detects all anomalies.
P3 (Age 8)	Instead of audio/visual alerts, students interpret live data charts with signal noise. They must identify hazard spikes and input precise grid coordinates to safely simulate construction.
P4 (Age 9)	Operating under a strict countdown timer, students strategically toggle between a slow "Normal Scan" and a fast but blurry "Turbo Scan" to map the entire grid efficiently.
P5 (Age 10)	Students analyze chart textures to differentiate harmless objects from hazards, using these visual patterns to train an AI model for automated hazard recognition.

P6 (Age 11)	Students manage limited "Drill Samples" to verify suspicious blurry data from Turbo Scans, finalizing construction blueprints. They also tune system sliders to balance execution speed and detection accuracy, applying these concepts to an automated airport luggage scanner simulation.
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Lesson 3: Grid Navigation & Algorithmic Pathfinding

Level	Description
P1 (Age 6)	Students provide step-by-step directional commands to guide a character from point A to point B on a grid.
P2 (Age 7)	Participants formulate "if-then" conditional rules to enable a bot to navigate a grid autonomously.
P3 (Age 8)	Students plan optimal paths with limited visibility, seeing only two squares ahead at any given time.
P4 (Age 9)	In a robot vacuum simulation, students map a cluttered environment and categorize obstacles into permanent fixtures and temporary obstructions.
P5 (Age 10)	Students program a robot vacuum to clean an entire floor plan efficiently without collisions or redundant coverage.
P6 (Age 11)	Students simulate an autonomous street sweeper, programming complex rules to navigate traffic laws, weather conditions, pedestrian interactions, and battery management.

Lesson 4: Autonomous Drone Routing

Level	Description
P1 (Age 6)	Students manually trace a path on a tablet grid, connecting waypoints while avoiding static cloud obstacles.
P2 (Age 7)	Students observe an AI-piloted drone that autonomously navigates around slowly moving clouds.
P3 (Age 8)	Students plan flight paths through environments with strict no-fly zones (e.g., buildings, sanctuaries). They manage a speed-versus-handling trade-off, where higher speeds widen the flight path and increase collision risks.
P4 (Age 9)	Drones autonomously adjust their speed based on environmental context: fast in clear airspace, medium in turbulent areas, and slow near obstacles.
P5 (Age 10)	The map transforms into a spatial node network. Students calculate and select the most efficient route by minimizing numerical "cost" values representing battery drain and congestion.
P6 (Age 11)	Students use a visual scripting interface to compile flight scripts using sequential command blocks. They design a city-wide delivery network, utilizing loops and conditionals to dynamically reroute around new obstacles and optimize multi-stop deliveries.

Lesson 5: Swarm Intelligence & Pathfinding

Level	Description
P1 (Age 6)	Students guide a "Mama Duck" leader drone across a touchscreen, while a flock of "Baby Duck" drones autonomously trails behind to return to base.

P2 (Age 7)	Students navigate a tight drone cluster through an obstacle course, managing "safety force-fields" to prevent mid-air collisions in narrow gaps.
P3 (Age 8)	Participants route a drone swarm across a map with dynamic, shifting no-fly zones (e.g., moving weather or birds), deploying live waypoints to herd the swarm safely.
P4 (Age 9)	Students manage a large swarm using visual sliders to adjust core flocking algorithms: Separation, Cohesion, and Alignment.
P5 (Age 10)	Students optimize agricultural drone deployment, balancing speed, cargo capacity, and strict harvesting time limits to prevent crop spoilage.
P6 (Age 11)	Acting as air traffic controllers, students program right-of-way rules and launch timings for multiple intersecting autonomous delivery drones to ensure seamless, collision-free operations.

Lesson 6: Delivery Routing & Loop Optimization

Level	Description
P1 (Age 6)	Students tap sequential collection points to guide a garbage truck, learning the necessity of closing the route loop by returning to the depot.
P2 (Age 7)	Students navigate a truck through a town grid with one-way streets, connecting all stops into a continuous, valid loop.

P3 (Age 8)	Participants experiment with route sequences to minimize total distance traveled, optimizing fuel consumption across multiple collection points.
P4 (Age 9)	Tackling the Traveling Salesman Problem, students map multi-stop routes and compare their manual solutions against optimal paths generated by AI algorithms.
P5 (Age 10)	Students coordinate routes under strict capacity constraints, factoring in weight penalties that reduce speed and increase battery drain as the truck fills up.
P6 (Age 11)	Using a visual Route Builder Timeline, students cluster dozens of pending collections into optimized, battery-efficient loops, later comparing their logistical strategies against AI-generated optimal solutions.

Lesson 7: Autonomous Transit Scheduling

Level	Description
P1 (Age 6)	Students manually match color-coded passengers to corresponding buses at stops, later automating the process using basic "if-then" logic rules.
P2 (Age 7)	Participants establish simple timetables for single and multi-bus loops, utilizing conditional rules to align buses with specific stops and schedules.
P3 (Age 8)	Students analyze passenger demand heatmaps, dynamically adjusting bus frequencies to match real-time crowd densities at various stops.

P4 (Age 9)	Utilizing AI predictive tools, students forecast passenger surges based on time and virtual city events, scheduling preventative transit to avoid overcrowding.
P5 (Age 10)	Students design intersecting bus networks, coordinating transfer timings and departure schedules to minimize passenger wait times and optimize system-wide flow.
P6 (Age 11)	Acting as AI dispatchers, students manage a live transit grid, dynamically rerouting buses and reassigning assets in real-time to mitigate breakdowns and roadblocks.
Lesson 8: Urban Healthcare & Emergency Response	
Level	Description
P1 (Age 6)	Students triage ER patients by dragging color-coded tags based on symptoms and assigning them to appropriate ward beds before frustration meters deplete.
P2 (Age 7)	Students train an AI model to automate patient sorting. They also manage a convenience store's inventory, stocking items based on AI-forecasted weather and environmental conditions.
P3 (Age 8)	Students sort emergency supplies from a central depot to neighborhood clinics, utilizing an AI Traffic Override System to bypass gridlock and prevent clinic stockouts.
P4 (Age 9)	Students manage hospital staff distribution, balancing patient surges and preventing employee burnout by deploying AI administrative helpers and scheduling breaks.

P5 (Age 10)	Acting as City Planners, students analyze decade-long demographic heatmaps to strategically place high-capacity hospitals or rapid-deployment tech clinics within a fixed budget.
P6 (Age 11)	Students manage a dual crisis (aging demographics and severe heatwaves), utilizing 30-year predictive models to allocate resources across hospitals, clinics, and urban cooling infrastructure, while executing AI load-shedding to protect critical health services.

Lesson 9: Smart Water Distribution Networks

Level	Description
P1 (Age 6)	Students design a basic water network by connecting pipes, towers, and houses, tapping to repair visual leaks.
P2 (Age 7)	Participants monitor network pressure gauges, adjusting virtual valves to maintain optimal water flow throughout a neighborhood.
P3 (Age 8)	Students deploy virtual acoustic sensors, using AI audio analysis to pinpoint hidden leaks and execute targeted, resource-efficient repairs.
P4 (Age 9)	Utilizing weather and population data, students predict daily water demand and calibrate pumping station outputs to balance supply and energy efficiency.
P5 (Age 10)	Students schedule pump operations across a complex reservoir network, shifting heavy energy usage to off-peak hours while maintaining emergency reserves.

P6 (Age 11)	Managing a city-wide smart water grid during a drought, students enforce tiered restrictions, utilize AI to detect illegal usage, and reroute reclaimed water to maintain civic satisfaction.
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Lesson 10: Intelligent Power Grid Management

Level	Description
P1 (Age 6)	Students route power from a central hub to edge devices, learning about voltage drop and the benefits of local AI processing (Edge Computing) over long-distance transmission.
P2 (Age 7)	Students balance municipal power against the massive, unpredictable energy spikes caused by AI Data Center training cycles, prioritizing citizen comfort versus compute progress.
P3 (Age 8)	Participants optimize the geographical placement of AI Data Centers and renewable energy sources, leveraging natural cooling (rivers/mountains) to minimize power consumption and infrastructure strain.
P4 (Age 9)	Students manage energy shortfalls by balancing battery reserves, fossil fuel backups, and AI compute throttling, eventually training an AI to automate these allocation decisions.
P5 (Age 10)	Acting as Digital Market Directors, students regulate dynamic pricing and set power thresholds to prevent corporate AI agents from monopolizing the grid during energy deficits.
P6 (Age 11)	During a cascading infrastructure crisis, students command an AI Grid Orchestrator, executing automated load-shedding protocols to sacrifice non-essential AI services and preserve critical physical infrastructure.

Lesson 11: AI-Powered Urban Monitoring

Level	Description
P1 (Age 6)	Students use an AI camera to locate a lost item in a crowded park by evaluating similarity scores, then train a new AI model to recognize the specific object.
P2 (Age 7)	Students utilize Monocular Depth Estimation to generate 3D depth maps from 2D cameras, setting buffer zones to warn tall vehicles of low-clearance bridges.
P3 (Age 8)	Participants apply Edge Detection algorithms (Wireframe Mode) to strip away visual noise, allowing AI to accurately count pedestrians at complex crosswalks.
P4 (Age 9)	Students run Semantic Segmentation on post-disaster drone footage, color-coding pixels (water, debris, power lines) to generate clean emergency response routes.
P5 (Age 10)	Students calibrate AI alert sensitivity at intersections and subway gates to balance safety (detecting tailgating or accidents) with minimizing false alarms.
P6 (Age 11)	Using Person Re-Identification (Re-ID), students track a missing individual across a city-wide camera network by matching clothing and feature vectors, bypassing the need for facial recognition.

Lesson 12: Adaptive Traffic Signal Optimization

Level	Description
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P1 (Age 6)	Students train an AI vision model by tagging moving vehicles (cars, buses, pedestrians) to establish lane priority rules at crosswalks.
P2 (Age 7)	Students use predictive AI to trigger "V2X Swarm Detours," preemptively clearing civilian traffic from an ambulance's future path.
P3 (Age 8)	Participants monitor subterranean Inductive Loop Sensor data via bar graphs, flushing traffic queues before intersections reach maximum capacity and gridlock.
P4 (Age 9)	Students identify "Disaster Signatures" (sudden speed drops) on live graphs, triggering emergency "Green Wave" protocols to clear bottlenecks instantly.
P5 (Age 10)	Students train an AI model on human decision-making, balancing traffic priorities based on dynamic variables like weather, time of day, and environmental conditions.
P6 (Age 11)	Students tune master AI Reward Dials (Speed, Safety, Eco) to manage the mass exodus of autonomous vehicles from a major stadium event, optimizing the city's algorithmic response.

Lesson 13: Predictive Traffic Flow & Emergency Routing

Level	Description
P1 (Age 6)	Students observe phantom traffic jams (shockwaves) and deploy Smart Dash Warnings to synchronize braking across trailing vehicles.

P2 (Age 7)	Participants manage backward-moving traffic waves caused by slow vehicles, clearing space ahead to allow emergency vehicles to enter the highway safely.
P3 (Age 8)	Students adjust traffic light durations based on translucent AI Forecast Paths, absorbing incoming traffic waves before they materialize into jams.
P4 (Age 9)	Acting as routing coordinators, students manipulate dynamic lane barriers based on 10-second predictive density graphs to divide and conquer impending shockwaves.
P5 (Age 10)	Students manage the "Echo Effect" of rerouting, using AI dashboards to adjust surrounding traffic lights and prevent secondary bottlenecks in residential zones.
P6 (Age 11)	Students deploy Eulerian Street-Level Optimization agents, comparing manual traffic management against a micro-agent network that autonomously dissipates shockwaves and clears emergency paths.

Lesson 14: AI-Assisted Air Traffic Control

Level	Description
P1 (Age 6)	Students manage aircraft proximity by adjusting speeds to prevent AI Safety Shields from overlapping and triggering collision warnings.
P2 (Age 7)	Participants manage strict runway landing gaps using ghost timeline markers, training an AI to automate speed adjustments and penalize inefficient routing.

P3 (Age 8)	Students command sweeping turns for commercial jets, using predictive Ghost Trails to account for momentum and prevent future path intersections.
P4 (Age 9)	Students configure AI Autopilot priority sliders (Fuel Efficiency, Collision Avoidance, Size-Comfort) to route diverse aircraft through dynamic atmospheric wind vectors.
P5 (Age 10)	Students tune AI clustering filters to highlight genuine "Danger Clusters" on a crowded radar without triggering excessive false-positive alerts.
P6 (Age 11)	Students engage a 4D Trajectory-Based AI Engine to collaboratively route multiple aircraft around a massive, shifting storm system by synchronizing speed and altitude adjustments.

Lesson 15: Computational Tokenomics & Resource Allocation

Level	Description
P1 (Age 6)	Students allocate limited AI compute tokens (Bronze vs. Gold) to generate park designs, learning the trade-off between resource cost and output fidelity.
P2 (Age 7)	Participants manage system load during peak citizen inquiries by adjusting a "Surge Pricing" dial, throttling non-urgent tasks to prevent hardware overheating.
P3 (Age 8)	Students balance a fixed daily token budget for a digital passport app, prioritizing resources across text, application, and image-processing requests.

P4 (Age 9)	Students manage a 100-token budget to train specialized AI models (Audio, Text, Image), strategically halting training to balance accuracy against token depletion.
P5 (Age 10)	Students evaluate and select government AI engines based on processing speed, query costs, and accuracy rates.
P6 (Age 11)	Students monitor a live data center dashboard, balancing server compute loads and core temperatures to prevent hardware crashes during intensive machine learning tasks.

Lesson 16: Civic Sentiment Analysis

Level	Description
P1 (Age 6)	Students categorize floating emotional word bubbles into positive or negative bins, teaching an AI the emotional weight of specific vocabulary.
P2 (Age 7)	Participants analyze mixed-sentiment sentences, applying a rulebook to assign positive, negative, or neutral numerical scores to specific keywords.
P3 (Age 8)	Students route live citizen feedback to the correct municipal department (Parks, Transit, Waste) based on their calculated sentiment scores.
P4 (Age 9)	Acting as AI customer service agents, students analyze citizen sentiment and select the most appropriate conversational tone (Enthusiastic, Robotic, or Apologetic).

P5 (Age 10)	Students monitor a dynamic civic Word Cloud, identifying emerging issues via color and size shifts, and routing critical data to relevant authorities.
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P6 (Age 11)	Acting as City Mayor, students utilize macro-level sentiment analysis of public feedback to prioritize municipal resources and maintain voter approval.
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Lesson 17: AI in Municipal Finance & Auditing

Level	Description
P1 (Age 6)	Students fund citizen improvement requests using a limited coin box, choosing between basic fixes (Bronze) and high-tech solutions (Gold) to manage municipal spending.
P2 (Age 7)	Participants route financial aid applications to appropriate AI audit models based on complexity to optimize government processing costs.
P3 (Age 8)	Students configure dynamic billing parameters for an AI document parser, scaling compute fees based on the page count and RAM usage of corporate uploads.
P4 (Age 9)	Students manage a chaotic influx of citizen support tickets, combining continuous AI data labeling with manual emergency budget overrides to prevent district infrastructure collapse.
P5 (Age 10)	Acting as Human-in-the-Loop Auditors, students verify contested corporate tax anomalies, resolve false positives, and deploy compute probes to track offshore wire transfers.

P6 (Age 11)	Students manipulate macroeconomic sliders (Tax Rates, Infrastructure Budget, AI Audit Intensity) to navigate economic shocks, utilizing AI sandboxes to forecast inflation and GDP impacts.
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Lesson 18: Capstone: Designing the AI City

Level	Description
P1 (Age 6)	Students participate in an age-appropriate, guided city-building simulation. They place foundational infrastructure, manage simple resource budgets, and observe how their earlier AI modules interact within a living 2D city grid.
P2 (Age 7)	Students participate in an age-appropriate, guided city-building simulation. They place foundational infrastructure, manage simple resource budgets, and observe how their earlier AI modules interact within a living 2D city grid.
P3 (Age 8)	Students participate in an age-appropriate, guided city-building simulation. They place foundational infrastructure, manage simple resource budgets, and observe how their earlier AI modules interact within a living 2D city grid.
P4 (Age 9)	Students design a comprehensive AI City within a set budget. An AI assistant acts as a conversational trade-off analyzer, evaluating infrastructure placement (e.g., warning against placing Data Centers near residential zones). They toggle between three distinct visualization layers (Power & Utility, Logistics & Waste, Social & Safety) to manage complex infrastructure.

P5 (Age 10)	Students design a comprehensive AI City within a set budget. An AI assistant acts as a conversational trade-off analyzer, evaluating infrastructure placement (e.g., warning against placing Data Centers near residential zones). They toggle between three distinct visualization layers (Power & Utility, Logistics & Waste, Social & Safety) to manage complex infrastructure.
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P6 (Age 11)	Students design a comprehensive AI City within a set budget. An AI assistant acts as a conversational trade-off analyzer, evaluating infrastructure placement (e.g., warning against placing Data Centers near residential zones). They toggle between three distinct visualization layers (Power & Utility, Logistics & Waste, Social & Safety) to manage complex infrastructure.
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Lesson 19: Capstone: AI City Optimization

Level	Description
P1 (Age 6)	Students engage in basic logic mapping of city services, connecting simple cause-and-effect relationships within their urban layouts.
P2 (Age 7)	Students engage in basic logic mapping of city services, connecting simple cause-and-effect relationships within their urban layouts.
P3 (Age 8)	Students engage in basic logic mapping of city services, connecting simple cause-and-effect relationships within their urban layouts.

P4 (Age 9)	Students construct a knowledge graph to optimize their city design. By connecting conceptual node blocks with directional relationship links, they map out complex city mechanics while operating under strict AI token budget constraints.
P5 (Age 10)	Students construct a knowledge graph to optimize their city design. By connecting conceptual node blocks with directional relationship links, they map out complex city mechanics while operating under strict AI token budget constraints.
P6 (Age 11)	Students construct a knowledge graph to optimize their city design. By connecting conceptual node blocks with directional relationship links, they map out complex city mechanics while operating under strict AI token budget constraints.

Lesson 20: Capstone: Live Macro-Simulation

Level	Description
P1 (Age 6)	Students observe real-time city functions and simple cause-and-effect loops, tapping individual assets to see basic micro-behaviors in action.
P2 (Age 7)	Students observe real-time city functions and simple cause-and-effect loops, tapping individual assets to see basic micro-behaviors in action.
P3 (Age 8)	Students observe real-time city functions and simple cause-and-effect loops, tapping individual assets to see basic micro-behaviors in action.

P4 (Age 9)	Students run a live macro-simulation across varying time scales (Real-Time, 1 Day, 100 Years). They manage long-term macro-trends like aging demographics, property value shifts, and 'Black Swan' events (e.g., heatwaves, floods) to test their city's resilience.
P5 (Age 10)	Students run a live macro-simulation across varying time scales (Real-Time, 1 Day, 100 Years). They manage long-term macro-trends like aging demographics, property value shifts, and 'Black Swan' events (e.g., heatwaves, floods) to test their city's resilience.
P6 (Age 11)	Students run a live macro-simulation across varying time scales (Real-Time, 1 Day, 100 Years). They manage long-term macro-trends like aging demographics, property value shifts, and 'Black Swan' events (e.g., heatwaves, floods) to test their city's resilience.

課程 1: 智慧廢物分類

級別	描述
P1 (6歲)	學生參與互動式數位分類練習，透過拖放機制將廢棄物歸類到適當的回收桶與一般垃圾桶。他們也會參與逐步揭曉的猜謎遊戲，辨識可回收物。
P2 (7歲)	學員根據複雜的非視覺屬性 (如重量、材質成分、質地) 對物體進行分類。他們同時將描述性的提示卡與對應的物體照片配對。
P3 (8歲)	學生使用平板電腦，圈選影像中的明顯特徵來訓練 AI 模型。他們學習物體偵測 (邊界框)、幾何/地標 (辨識角落與稜線)、表面紋理分析，以及文字/符號辨識 (條碼與標誌)。
P4 (9歲)	奠基於先前的標註技能，學生根據特定特徵組合引導 AI 產生影像。他們也參與猜謎推理遊戲，僅從描述的特徵中辨識物體。
P5 (10歲)	學生將特徵標註遊戲應用於可回收材料，上傳標記好的資料集以訓練基礎的 AI 分類模型。
P6 (11歲)	學生使用可回收材料精進其 AI 分類模型，並學習機器學習中重要的「模型過度配適」概念。

課程 2: 地下 AI 掃描與場址分析

級別	描述
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P1 (6歲)	一台 AI 探測車繪製 2D 土地網格，以識別地下危險物 (例如管線、古物、地下水)。偵測器發出危險信號，標記「禁建區」，讓學生進行鑽探與調查。
P2 (7歲)	學生以兩種模式操作掃描器：快速掃描會遺漏深層且狹窄的危險物；慢速高頻脈衝則可偵測所有異常。
P3 (8歲)	學生不依靠聲音或視覺警報，而是解讀即時數據圖表中的訊號雜訊。他們必須識別危險數據峰值，並輸入精確的網格座標以安全模擬施工。
P4 (9歲)	在嚴格的倒數計時下，學生策略性地切換緩慢的「正常掃描」與快速但模糊的「渦輪掃描」，以高效繪製整個網格。
P5 (10歲)	學生分析圖表紋理，區分無害物體與危險物，並利用這些視覺模式訓練 AI 模型來自動辨識危險物。
P6 (11歲)	學生管理有限的「鑽探樣本」，驗證渦輪掃描中可疑的模糊數據，最終完成施工藍圖。他們也調整系統滑桿以平衡執行速度與偵測準確度，並將這些概念應用於自動化機場行李掃描模擬。

課程 3：網格導航與演算法路徑規劃

級別	描述
P1 (6歲)	學生提供逐步的方向指令，引導角色在網格上從 A 點移動到 B 點。
P2 (7歲)	學員制定「若則」條件規則，使機器人能夠自主導航網格。
P3 (8歲)	學生在能見度有限的條件下規劃最佳路徑，任何時間僅能看見前方兩格。

P4(9歲)	在機器人吸塵器模擬中，學生繪製雜亂環境的地圖，並將障礙物分類為永久固定物與臨時阻礙物。
P5(10歲)	學生為機器人吸塵器編寫程式，使其能有效清潔整個樓層平面，避免碰撞與重複覆蓋。
P6(11歲)	學生模擬自動化街道清掃車，編寫複雜的規則來應對交通法規、天氣條件、行人互動以及電池管理。
課程 4: 自主無人機航線規劃	
級別	描述
P1(6歲)	學生在平板網格上手動描繪路徑，連接航點，同時避開靜態雲朵障礙物。
P2(7歲)	學生觀察一架 AI 駕駛的無人機，自主繞過緩慢移動的雲朵。
P3(8歲)	學生規劃穿越嚴格禁航區(例如建築物、保護區)的飛行路徑。他們管理速度與操控性的取捨:速度越快，飛行路徑越寬，碰撞風險越高。
P4(9歲)	無人機根據環境情境自動調整速度:在淨空空域快速飛行，在亂流區域中速飛行，在接近障礙物時慢速飛行。
P5(10歲)	地圖轉變為空間節點網路。學生計算並選擇最有效率的路線，最小化代表電池消耗與擁擠程度的數值「成本」。
P6(11歲)	學生使用視覺化腳本介面，以循序指令方塊編寫飛行腳本。他們設計一個城市級配送網路，利用迴圈與條件判斷動態繞開新障礙物，並最佳化多站點配送。

課程 5: 群體智慧與路徑規劃

級別	描述
P1 (6歲)	學生在觸控螢幕上引導「鴨媽媽」領頭無人機, 而一群「鴨寶寶」無人機則自主跟隨在後, 最終返回基地。
P2 (7歲)	學生帶領緊密的無人機集群穿越障礙賽道, 管理「安全力場」以防止在狹窄間隙中發生空中碰撞。
P3 (8歲)	學員引導無人機集群穿越具有動態、移動式禁航區(例如移動中的天氣系統或鳥群)的地圖, 部署即時航點以安全地引導集群。
P4 (9歲)	學生使用視覺滑桿調整核心群體演算法的參數: 分離、cohesion (凝聚)、alignment (對齊)。
P5 (10歲)	學生最佳化農業無人機部署, 平衡速度、載貨容量與嚴格的採收時間限制, 以防止作物腐壞。
P6 (11歲)	學生擔任飛航管制員, 為多架相互交錯的自主配送無人機編寫路權規則與起飛時序, 確保無縫且無碰撞的運作。

課程 6: 配送路徑與迴圈最佳化

級別	描述
P1 (6歲)	學生依序點擊收集點來引導垃圾車, 學習路線必須返回調度場以形成封閉迴圈的必要性。
P2 (7歲)	學生駕駛垃圾車穿越具有單行道的城鎮網格, 將所有停靠點連接成一個連續且有效的迴圈。

P3 (8歲)	學員嘗試不同的路線順序, 以最小化總行駛距離, 在多個收集點之間最佳化燃料消耗。
P4 (9歲)	挑戰旅行推銷員問題, 學生繪製多站點路線, 並將其手動解決方案與 AI 演算法生成的最佳路徑進行比較。
P5 (10歲)	學生在嚴格的容量限制下協調路線, 考慮重量懲罰(隨著卡車載重增加, 速度下降且電池消耗增加)。
P6 (11歲)	學生使用視覺化路線建立時間軸, 將數十個待處理的收集點集群成最佳化且省電的迴圈, 之後再將其物流策略與 AI 生成的最佳解決方案進行比較。

課程 7: 自動化公共運輸排程

級別	描述
P1 (6歲)	學生手動將顏色標記的乘客與對應的公車站牌配對, 之後再利用基本的「若則」邏輯規則自動化此過程。
P2 (7歲)	學員為單一及多條公車迴圈建立簡單的時間表, 利用條件規則使公車與特定站牌及班表對齊。
P3 (8歲)	學生分析乘客需求熱區圖, 動態調整公車頻率以因應各站點的即時人群密度。
P4 (9歲)	學生運用 AI 預測工具, 根據時間與虛擬城市活動預測乘客激增時段, 安排預防性運輸以避免過度擁擠。
P5 (10歲)	學生設計交錯的公車網路, 協調轉乘時間與發車班表, 以最小化乘客等待時間並最佳化整體系統流量。

P6(11歲)	學生擔任 AI 派遣員，管理即時運輸網路，即時動態重新規劃公車路線並重新調度資產，以緩解車輛故障與道路封閉的影響。
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課程 8: 都市醫療與緊急應變

級別	描述
P1(6歲)	學生根據症狀拖曳顏色標籤來為急診病患檢傷分類，並將其分配到適當的病房床位，在挫折感計量表耗盡前完成。
P2(7歲)	學生訓練 AI 模型來自動化病患分類。他們也管理便利商店庫存，根據 AI 預測的天氣與環境條件進貨。
P3(8歲)	學生將緊急物資從中央倉庫分配到鄰里診所，運用 AI 交通優先系統繞過壅塞，防止診所庫存耗盡。
P4(9歲)	學生管理醫院人力分配，平衡病患激增與員工過勞，部署 AI 行政助手並安排休息時間。
P5(10歲)	學生擔任城市規劃師，分析長達十年的人口熱區圖，在固定預算內策略性地設置大型醫院或快速部署技術診所。
P6(11歲)	學生管理雙重危機(人口老化與嚴重的熱浪)，利用 30 年預測模型將資源分配給醫院、診所與都市降溫基礎設施，同時執行 AI 卸載保護關鍵醫療服務。

課程 9: 智慧水資源分配網路

級別	描述
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P1 (6歲)	學生連接水管、水塔與房屋來設計基本水網路，點擊修復可見的漏水。
P2 (7歲)	學員監控網路壓力計，調整虛擬閥門以維持整個社區的最佳水流。
P3 (8歲)	學生部署虛擬聲學感測器，利用 AI 音訊分析精確定位隱藏的漏水點，並執行針對性、資源效率高的修復。
P4 (9歲)	利用天氣與人口數據，學生預測每日用水需求，並校準抽水站輸出，以平衡供應與能源效率。
P5 (10歲)	學生在複雜的水庫網路中安排抽水機運作，將高耗能作業轉移到離峰時段，同時維持緊急備用水源。
P6 (11歲)	學生在乾旱期間管理全市智慧水網，實施分級限水措施，利用 AI 偵測非法用水，並重新引導回收水以維持市民滿意度。

課程 10: 智慧電網管理

級別	描述
P1 (6歲)	學生將電力從中央樞紐輸送到邊緣裝置，學習電壓降以及本地 AI 處理(邊緣運算)相較於長距離傳輸的優勢。
P2 (7歲)	學生平衡市政用電與 AI 資料中心訓練週期所產生的大規模、不可預測的能源尖峰，在市民舒適度與運算進度之間取捨。
P3 (8歲)	學員最佳化 AI 資料中心與再生能源的地理位置，利用自然冷卻(河流/山區)以最小化耗電量與基礎設施負擔。

P4(9歲)	學生透過平衡電池儲能、化石燃料備用以及 AI 運算限流來管理能源短缺，最終訓練一個 AI 來自動化這些分配決策。
P5(10歲)	學生擔任數位市場主管，監管動態定價並設定電力閾值，防止企業 AI 代理人在能源短缺期間壟斷電網。
P6(11歲)	在連鎖基礎設施危機期間，學生指揮 AI 電網調度系統，執行自動卸載協議，犧牲非必要的 AI 服務以保護關鍵實體基礎設施。

課程 11:AI 城市監控

級別	描述
P1(6歲)	學生使用 AI 攝影機在擁擠的公園中透過評估相似度分數尋找遺失物品，然後訓練新的 AI 模型來辨識該特定物體。
P2(7歲)	學生運用單眼深度估計，從 2D 攝影機生成 3D 深度圖，設定緩衝區以警告高車輛注意低淨空橋梁。
P3(8歲)	學員應用邊緣偵測演算法(線框模式)去除視覺雜訊，讓 AI 能夠準確計算複雜人行穿越道上的行人數量。
P4(9歲)	學生在災後無人機空拍畫面上執行語義分割，對像素進行顏色編碼(水體、瓦礫、電線)，以產生乾淨的緊急應變路線。
P5(10歲)	學生校準路口與地鐵閘門的 AI 警示敏感度，以平衡安全(偵測跟隨闖入或事故)與減少誤報。
P6(11歲)	學生運用行人再辨識技術，透過比對衣著與特徵向量，在整個城市的攝影機網路中追蹤失蹤者，無需使用臉部辨識。

課程 12:適應性交通號誌最佳化

級別	描述
P1 (6歲)	學生透過標記移動中的車輛(汽車、公車、行人)訓練 AI 視覺模型, 以建立行人穿越道的車道優先規則。
P2 (7歲)	學生使用預測性 AI 觸發「V2X 群體繞道」, 預先清除救護車未來路徑上的民用車流。
P3 (8歲)	學員透過長條圖監控地下感應迴路線圈數據, 在路口達到最大容量與死結前清空排隊車流。
P4 (9歲)	學生在即時圖表上識別「災難特徵」(速度驟降), 立即觸發緊急「綠波」協定以清除瓶頸。
P5 (10歲)	學生訓練 AI 模型模仿人類決策, 根據天氣、時間、環境條件等動態變數平衡交通優先權。
P6 (11歲)	學生調整主 AI 獎勵轉盤(速度、安全、環保), 管理大型體育場活動結束後大量自動駕駛車輛的疏散, 最佳化城市的演算法回應。

課程 13:預測性交通流與緊急路線規劃

級別	描述
P1 (6歲)	學生觀察幽靈交通壅塞(衝擊波), 並部署智慧車載預警系統以同步後方車輛的煞車。
P2 (7歲)	學員管理由慢速車輛引起的後向移動交通波, 清理前方空間讓緊急車輛能安全進入高速公路。

P3(8歲)	學生根據半透明的 AI 預測路徑調整紅綠燈秒數，在交通波形成壅塞前將其吸收。
P4(9歲)	學生擔任路線協調員，根據 10 秒預測密度圖操作動態車道屏障，以分流即將到來的衝擊波。
P5(10歲)	學生管理路線改道的「回音效應」，利用 AI 儀表板調整周圍的交通號誌，防止住宅區形成二次瓶頸。
P6(11歲)	學生部署歐拉級街區最佳化代理程式，比較手動交通管理與微代理網路（自主消散衝擊波並清除緊急路徑）的成效。

課程 14: AI 輔助飛航管制

級別	描述
P1(6歲)	學生透過調整速度來管理飛機之間的距離，防止 AI 安全防護罩重疊並觸發碰撞警告。
P2(7歲)	學員使用重影時間軸標記嚴格管理跑道降落間隔，訓練 AI 自動化速度調整，並對無效率的路線予以懲罰。
P3(8歲)	學生指揮商業客機進行大轉彎，利用預測性重影軌跡考慮動量，防止未來路徑交錯。
P4(9歲)	學生設定 AI 自動駕駛優先權滑桿（燃油效率、防撞、體型與舒適度），引導多樣化的飛機通過動態大氣風向量。
P5(10歲)	學生調整 AI 分群濾波器，在擁擠的雷達螢幕上凸顯真正的「危險群集」，而不觸發過多的假警報。
P6(11歲)	學生啟動基於 4D 軌跡的 AI 引擎，協同引導多架飛機繞過大範圍移動中的風暴系統，同步調整速度與高度。

課程 15: 計算代幣經濟學與資源分配

級別	描述
P1 (6歲)	學生分配有限的 AI 計算代幣 (青銅 vs. 黃金) 來生成公園設計, 學習資源成本與輸出品質之間的取捨。
P2 (7歲)	學員在市民查詢尖峰時段管理系統負載, 調整「動態定價」轉盤, 限制非緊急任務以防止硬體過熱。
P3 (8歲)	學生為數位護照應用程式平衡固定的每日代幣預算, 優先分配資源給文字、應用程式和影像處理請求。
P4 (9歲)	學生管理 100 枚代幣的預算來訓練專門的 AI 模型 (音訊、文字、影像), 策略性地停止訓練以平衡準確度與代幣消耗。
P5 (10歲)	學生根據處理速度、查詢成本與準確率評估並選擇政府 AI 引擎。
P6 (11歲)	學生監控即時數據中心儀表板, 平衡伺服器運算負載與核心溫度, 防止在密集機器學習任務期間硬體當機。

課程 16: 公民情緒分析

級別	描述
P1 (6歲)	學生將漂浮的情緒文字氣泡分類為正面或負面, 教導 AI 特定詞彙的情緒權重。
P2 (7歲)	學員分析混合情緒的句子, 應用規則手冊為特定關鍵字指派正面、負面或中性的數值分數。

P3(8歲)	學生根據計算出的情緒分數, 將即時市民反饋引導至正確的市政部門(公園、交通、廢棄物管理)。
P4(9歲)	學生擔任 AI 客服人員, 分析市民情緒並選擇最合適的對話語氣(熱情、機器式或抱歉)。
P5(10歲)	學生監控動態的市政文字雲, 透過顏色與大小變化識別新興議題, 並將關鍵數據傳送給相關主管機關。
P6(11歲)	學生擔任市長, 運用公眾反饋的宏觀情緒分析來優先分配市政資源, 並維持民眾支持度。

課程 17: 市政財務與稽核中的 AI

級別	描述
P1(6歲)	學生使用有限的錢幣箱資助市民改善請求, 在基本修繕(青銅)與高科技解決方案(黃金)之間選擇, 以管理市政支出。
P2(7歲)	學員根據複雜度將財務援助申請轉介給適當的 AI 稽核模型, 以最佳化政府處理成本。
P3(8歲)	學生為 AI 文件解析器設定動態計費參數, 根據企業上傳的頁面數與 RAM 使用量調整計算費用。
P4(9歲)	學生管理混亂湧入的市民支援單, 結合連續的 AI 數據標記與手動緊急預算超支, 防止區域基礎設施崩潰。
P5(10歲)	學生擔任人機協同稽核員, 驗證有爭議的企業稅務異常, 解決誤報, 並部署計算探針追蹤離岸電匯。

P6(11歲)	學生操作宏觀經濟滑桿(稅率、基礎設施預算、AI 稽核強度)以應對經濟衝擊, 利用 AI 沙盒預測通膨與 GDP 影響。
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課程 18:總整課程:設計 AI 城市

級別	描述
P1(6歲)	學生參與適齡的引導式城市建設模擬。他們放置基礎建設, 管理簡單的資源預算, 並觀察先前學過的 AI 模組如何在一個生動的 2D 城市網格中互動。
P2(7歲)	學生參與適齡的引導式城市建設模擬。他們放置基礎建設, 管理簡單的資源預算, 並觀察先前學過的 AI 模組如何在一個生動的 2D 城市網格中互動。
P3(8歲)	學生參與適齡的引導式城市建設模擬。他們放置基礎建設, 管理簡單的資源預算, 並觀察先前學過的 AI 模組如何在一個生動的 2D 城市網格中互動。
P4(9歲)	學生在固定預算內設計全面的 AI 城市。一位 AI 助理擔任對話式的權衡分析師, 評估基礎設施的放置(例如警告不要將資料中心放在住宅區附近)。他們在三種不同的視覺化圖層(電力與公用事業、物流與廢棄物、社會與安全)之間切換, 以管理複雜的基礎設施。
P5(10歲)	學生在固定預算內設計全面的 AI 城市。一位 AI 助理擔任對話式的權衡分析師, 評估基礎設施的放置(例如警告不要將資料中心放在住宅區附近)。他們在三種不同的視覺化圖層(電力與公用事業、物流與廢棄物、社會與安全)之間切換, 以管理複雜的基礎設施。

P6(11歲)	學生在固定預算內設計全面的 AI 城市。一位 AI 助理擔任對話式的權衡分析師，評估基礎設施的放置(例如警告不要將資料中心放在住宅區附近)。他們在三種不同的視覺化圖層(電力與公用事業、物流與廢棄物、社會與安全)之間切換，以管理複雜的基礎設施。
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課程 19:總整課程:AI 城市最佳化

級別	描述
P1(6歲)	學生進行城市服務的基礎邏輯映射，在其城市布局中連接簡單的因果關係。
P2(7歲)	學生進行城市服務的基礎邏輯映射，在其城市布局中連接簡單的因果關係。
P3(8歲)	學生進行城市服務的基礎邏輯映射，在其城市布局中連接簡單的因果關係。
P4(9歲)	學生建立知識圖譜以最佳化其城市設計。透過連接概念節點區塊與有向關係鏈接，他們繪製複雜的城市機制，同時在嚴格的 AI 代幣預算限制下操作。
P5(10歲)	學生建立知識圖譜以最佳化其城市設計。透過連接概念節點區塊與有向關係鏈接，他們繪製複雜的城市機制，同時在嚴格的 AI 代幣預算限制下操作。
P6(11歲)	學生建立知識圖譜以最佳化其城市設計。透過連接概念節點區塊與有向關係鏈接，他們繪製複雜的城市機制，同時在嚴格的 AI 代幣預算限制下操作。

課程 20:總整課程:即時宏觀模擬

級別	描述
P1(6歲)	學生觀察即時城市功能與簡單的因果循環，點擊個別資產以查看基本的微觀行為。
P2(7歲)	學生觀察即時城市功能與簡單的因果循環，點擊個別資產以查看基本的微觀行為。
P3(8歲)	學生觀察即時城市功能與簡單的因果循環，點擊個別資產以查看基本的微觀行為。
P4(9歲)	學生在不同時間尺度(即時、1天、100年)下執行即時宏觀模擬。他們管理長期的宏觀趨勢，如人口老化、房產價值變動，以及「黑天鵝」事件(例如熱浪、洪水)，以測試其城市的韌性。
P5(10歲)	學生在不同時間尺度(即時、1天、100年)下執行即時宏觀模擬。他們管理長期的宏觀趨勢，如人口老化、房產價值變動，以及「黑天鵝」事件(例如熱浪、洪水)，以測試其城市的韌性。
P6(11歲)	學生在不同時間尺度(即時、1天、100年)下執行即時宏觀模擬。他們管理長期的宏觀趨勢，如人口老化、房產價值變動，以及「黑天鵝」事件(例如熱浪、洪水)，以測試其城市的韌性。